Bernoulli Distribution

1 Definition

Consider a random experiment whose sample space has been divided into two categories (success/failure). Then, any performance of the random experiment is known as Bernoulli trail. (Trail is an attempt to produce event, the event is neither impossible or certain.)

Let X denotes the outcome of a Bernoulli trail such that

$$X = \begin{cases} 1 & \text{if success occurs} \\ 0 & \text{if failure occurs} \end{cases}$$

Here X is a Bernoulli random variable, $p(\cdot) = p$.

So, $X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } (1-p) = q \end{cases}$

The pmf of X is

$$f_x(x) = \begin{cases} p^x(1-p)^{1-x}, & x = 0, 1 \\ 0, & \text{otherwise} \end{cases}$$

2 Moments

1. $E(X) = p$
2. $V(X) = p - p^2 = p(1-p) = pq \leqslant \frac{1}{4}$

   As, $AM \geqslant GM$
   $\frac{p+q}{2} \geqslant \sqrt{pq}$
   $\frac{1}{2} \geqslant \sqrt{pq}$
   $pq \leqslant \frac{1}{4}$

3 Problems

1. If X is Bernoulli random variable, $P(X=0) = 1 - P(X=1)$ and $E(X) = 3V(X)$. Find $P(X=0)$.

$$\begin{aligned} & E(X) = 3V(X) \\ \Rightarrow & p = 3p(1-p) = 3p - 3p^2 \\ \Rightarrow & 2p - 3p^2 = 0 \\ \Rightarrow & p(2-3p) = 0 \\ \Rightarrow & p = 0, \quad p = 2/3 \end{aligned}$$

$$P(X=0) = p^0(1-p)^1 = (1-p) = \begin{cases} 1, & p = 0 \\ \frac{1}{3}, & p = 2/3 \end{cases}$$

2. Let X have pdf $f_X(x) = \dfrac{1}{\pi\,(1+x^2)}, x \in \mathbb{R}$.

$$Y = \begin{cases} 1 & |X| \geqslant 1 \\ 0 & \text{otherwise} \end{cases}$$ Find the distribution of Y .

$$\begin{aligned} P(Y=1) &= P(|X| \geqslant 1) \\ &= 1 - P(|X| < 1) \\ &= 1 - P(-1 < X < 1) \\ &= 1 - \int_{-1}^{1} \frac{1}{\pi} \frac{1}{(1+x^2)} dx \\ &= 1 - \frac{1}{\pi} \left[\tan^{-1} x\right]_{-1}^{1} \\ &= 1 - \frac{2}{\pi} \tan^{-1}(1) \\ &= 1 - \frac{2}{\pi}\frac{\pi}{4} = 1 - \frac{1}{2} = \frac{1}{2} \end{aligned}$$

$$Y = \begin{cases} 1 & p = 1/2 \\ 0 & q = 1/2 \end{cases}$$

$Y \sim$ Bernoulli $\left(p = \frac{1}{2}\right)$

4 Two Point Distribution :

$$X = \begin{cases} a & \text{with probability } p \\ b & \text{with probability } q = 1-p \end{cases}$$

then, X is a random variable having two point distribution.

$$Y = \frac{X-b}{a-b} = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } q \end{cases}$$

then, Y is a Bernoulli random variable.

Every two point distribution can be converted into Bernoulli Distribution.